

O'Neill §1.3 Problem Set

Exercise 1. Let $\mathbf{v}_{\mathbf{p}}$ be the tangent vector to $\mathbf{E}^3$ for which $\mathbf{v} = (2, -1, 3)$ and $\mathbf{p} = (2, 0, -1)$. Working directly from the definition, compute the directional derivative $\mathbf{v}_{\mathbf{p}}[f]$, where

(a) $f = y^2z$.

(b) $f = x^7$.

(c) $f = e^x \cos y$.

Solution. By definition, the directional derivative is computed via the limit:

$$\mathbf{v}_{\mathbf{p}}[f] = \frac{d}{dt} \left(f(\mathbf{p} + t\mathbf{v}) \right) \Big|_{t=0}$$

The curve is given by $\mathbf{p} + t\mathbf{v} = (2 + 2t, -t, -1 + 3t)$.

(a) For $f = y^2z$:

$$f(\mathbf{p} + t\mathbf{v}) = (-t)^2(-1 + 3t) = -t^2 + 3t^3$$

Taking the derivative with respect to t at $t = 0$:

$$\frac{d}{dt}(-t^2 + 3t^3) \Big|_{t=0} = (-2t + 9t^2) \Big|_{t=0} = 0.$$

(b) For $f = x^7$:

$$f(\mathbf{p} + t\mathbf{v}) = (2 + 2t)^7$$

Taking the derivative with respect to t at $t = 0$:

$$\frac{d}{dt}(2 + 2t)^7 \Big|_{t=0} = 7(2 + 2t)^6 \cdot 2 \Big|_{t=0} = 14(2)^6 = 896.$$

(c) For $f = e^x \cos y$:

$$f(\mathbf{p} + t\mathbf{v}) = e^{2+2t} \cos(-t) = e^{2+2t} \cos t$$

Applying the product rule and evaluating at $t = 0$:

$$\frac{d}{dt} (e^{2+2t} \cos t) \Big|_{t=0} = (2e^{2+2t} \cos t - e^{2+2t} \sin t) \Big|_{t=0} = 2e^2(1) - e^2(0) = 2e^2.$$

□

Exercise 2. Let $V = y^2U_1 - xU_3$, and let $f = xy, g = z^3$. Compute the functions:

(a) $V[f]$.

(b) $V[g]$.

(c) $V[fg]$.

(d) $fV[g] - gV[f]$.

(e) $V[f^2 + g^2]$.

(f) $V[V[f]]$.

Solution. Recall that $V[h] = v_1 \frac{\partial h}{\partial x} + v_2 \frac{\partial h}{\partial y} + v_3 \frac{\partial h}{\partial z} = y^2 \frac{\partial h}{\partial x} - x \frac{\partial h}{\partial z}$.

(a) $V[f] = y^2 \frac{\partial(xy)}{\partial x} - x \frac{\partial(xy)}{\partial z} = y^2(y) - x(0) = y^3$.

(b) $V[g] = y^2 \frac{\partial(z^3)}{\partial x} - x \frac{\partial(z^3)}{\partial z} = y^2(0) - x(3z^2) = -3xz^2$.

(c) By the product rule for vector fields:

$$V[fg] = V[f]g + fV[g] = (y^3)(z^3) + (xy)(-3xz^2) = y^3z^3 - 3x^2yz^2.$$

(d) Substituting our answers from parts (a) and (b):

$$fV[g] - gV[f] = (xy)(-3xz^2) - (z^3)(y^3) = -3x^2yz^2 - y^3z^3.$$

(e) By linearity and the chain rule:

$$V[f^2 + g^2] = 2fV[f] + 2gV[g] = 2(xy)(y^3) + 2(z^3)(-3xz^2) = 2xy^4 - 6xz^5.$$

(f) Applying V to the resulting function from part (a):

$$V[V[f]] = V[y^3] = y^2 \frac{\partial(y^3)}{\partial x} - x \frac{\partial(y^3)}{\partial z} = y^2(0) - x(0) = 0.$$

□

Exercise 3. Prove the identity $V = \sum V[x_i]U_i$, where x_1, x_2, x_3 are the natural coordinate functions. (Hint: Evaluate $V = \sum v_i U_i$ on x_j .)

Solution. Let the vector field V be expressed in its standard form $V = \sum_{i=1}^3 v_i U_i$. We evaluate the action of V on an arbitrary natural coordinate function x_j (where $j = 1, 2, 3$):

$$V[x_j] = \left(\sum_{i=1}^3 v_i U_i \right) [x_j] = \sum_{i=1}^3 v_i U_i[x_j]$$

By definition of the natural frame fields, $U_i[x_j] = \frac{\partial x_j}{\partial x_i} = \delta_{ij}$, where δ_{ij} is the Kronecker delta symbol. Thus, the summation collapses to a single term:

$$V[x_j] = \sum_{i=1}^3 v_i \delta_{ij} = v_j$$

Since $v_j = V[x_j]$ for each component $j = 1, 2, 3$, we substitute these coefficient functions back into the standard form of V :

$$V = \sum_{i=1}^3 v_i U_i = \sum_{i=1}^3 V[x_i] U_i$$

This completes the proof of the identity. □

Exercise 4. If $V[f] = W[f]$ for every function f on $\mathbf{E}^3$, prove that $V = W$.

Solution. Let $V = \sum v_i U_i$ and $W = \sum w_i U_i$ be two vector fields on $\mathbf{E}^3$. We are given that $V[f] = W[f]$ for any real-valued function f .

Let us choose f to be the specific natural coordinate functions $f = x_j$ for $j = 1, 2, 3$. Substituting x_j into our given condition yields:

$$V[x_j] = W[x_j]$$

Using the identity proven in Exercise 4 (or evaluating directly via $U_i[x_j] = \delta_{ij}$), we know that $V[x_j] = v_j$ and $W[x_j] = w_j$. Thus, it follows directly that:

$$v_j = w_j \quad \text{for } j = 1, 2, 3$$

Since all coefficient functions are identically equal at every point, the vector fields themselves must be equal:

$$V = \sum v_i U_i = \sum w_i U_i = W$$

Thus, $V = W$. □